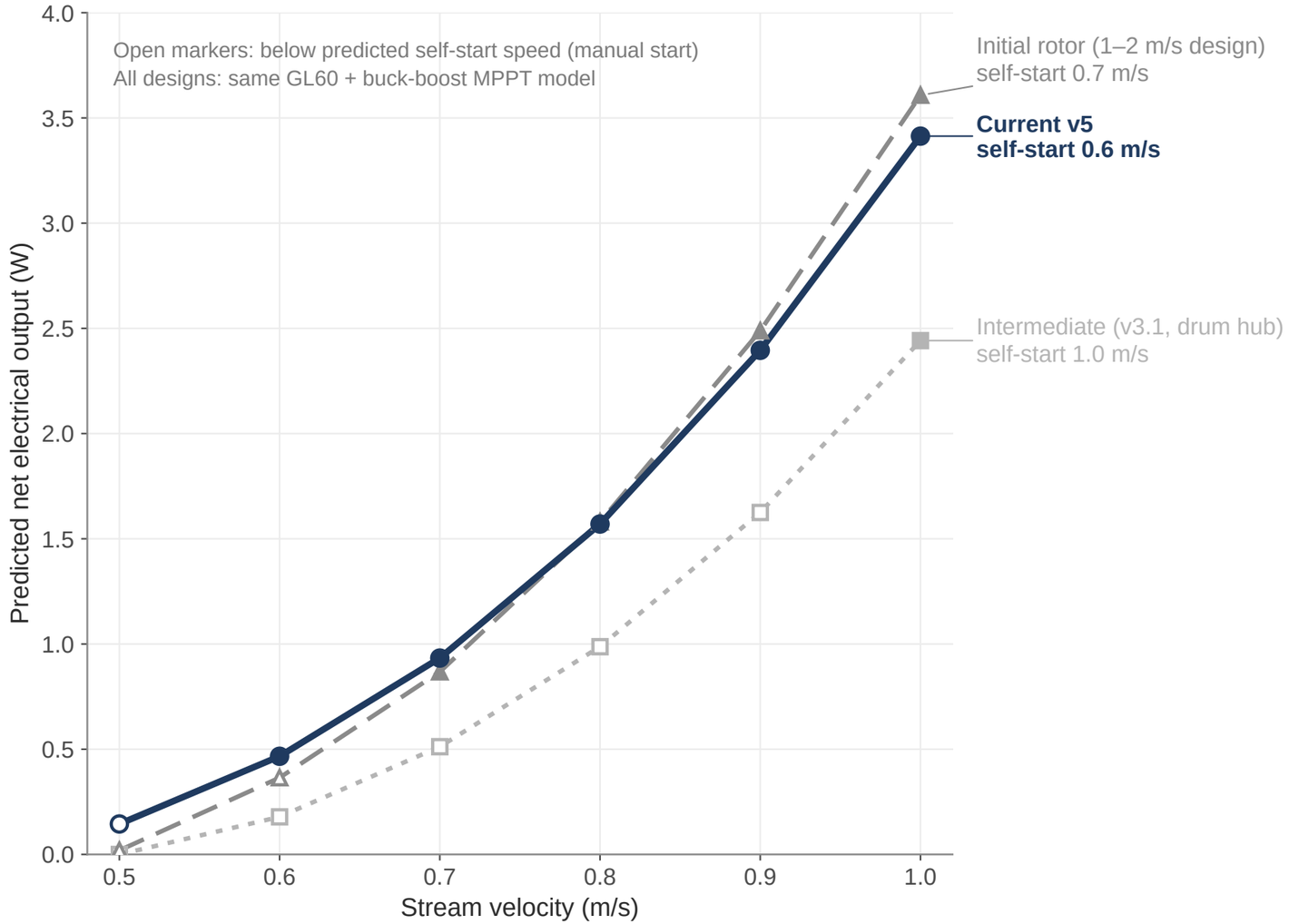
Turbine Optimization & Generator Matching

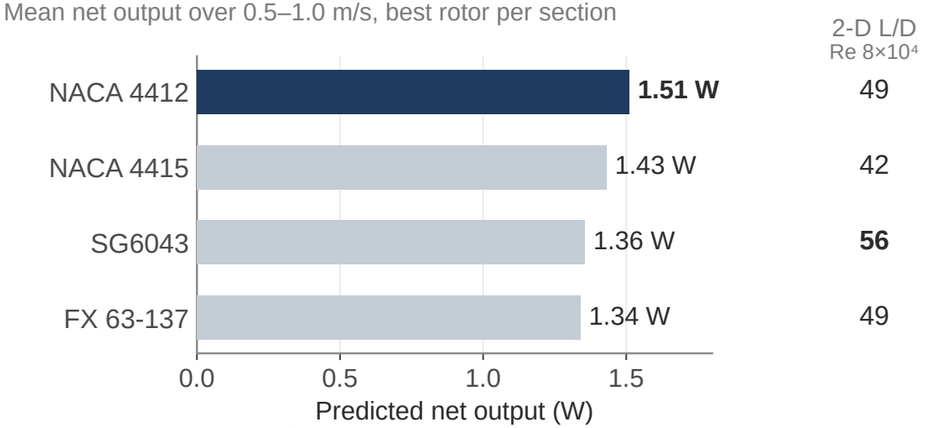
Low-flow refinement raises predicted output at 0.5–0.7 m/s

Current v5 vs initial rotor: +0.10 W (28 %) at 0.6 m/s; self-start 0.6 vs 0.7 m/s



NACA 4412 gives the highest predicted system output

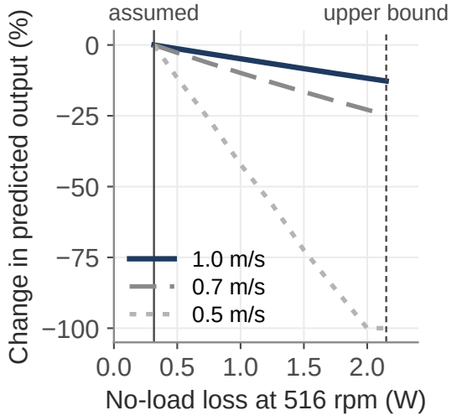
Mean net output over 0.5–1.0 m/s, best rotor per section



Airfoil polars: NeuralFoil (XFOIL surrogate), not measured.

GL60 KV25: no-load loss is the main open uncertainty

Upper bound: manufacturer no-load current (may include driver draw)



MANUFACTURER DATA	
Rated torque	0.60 N·m
Rated current	1.35 A
Speed constant	21.5 rpm/V
Torque constant	0.44–0.48 N·m/A
PREDICTED, CURRENT V5, 0.5–1.0 m/s	
Operating torque	0.03–0.16 N·m
Share of rated torque	4–26 %
ASSUMED, UNVERIFIED	
No-load loss, 516 rpm	0.32 W
Cogging torque	24 mN·m

Model predictions; experimental validation pending